

Andrés Carvajal

Atlanta, GA

Available Immediately | Open to Remote U.S. Contract & Full-Time Roles

andres_csuares@live.com — (470) 926-3159

[LinkedIn](#) — [Portfolio](#) — [GitHub](#)

DATA & ANALYTICS ENGINEER

SUMMARY

Data & Analytics Engineer with 3+ years of experience building production **data pipelines**, backend services, analytics workflows, and cloud-based systems across legal, commerce, healthcare, media, and industrial domains. Hands-on with **Python**, **SQL**, **Pandas**, **Polars**, **PostgreSQL**, **BigQuery**, **ETL**, and **AWS/GCP**. Strong at turning structured and unstructured data into reliable production workflows, from document extraction and retrieval to dashboards, evaluation pipelines, and automation services.

EXPERIENCE

RX-M

Sep. 2025 - Mar. 2026

Data & AI Engineer / Consultant

Remote, USA

- Developed **Python** and **FastAPI** services that orchestrated parsing, validation, retries, and artifact generation across multi-step production workflows.
- Designed canonical **data and retrieval pipelines** using **PostgreSQL**, **Pinecone**, **Qdrant**, and local **BM25** search to ground outputs against domain-specific source material.
- Integrated **OpenAI**, **Claude**, and **Gemini** APIs with **AWS** and **GCP** services while building evaluation and debug-artifact workflows that improved traceability and production reliability.

RandomAI

Jan. 2025 - Sep. 2025

Data & AI Engineer

Remote, USA

- Architected a **serverless platform** on **Firebase** and **GCP** that combined API services, data processing, storage, and event-driven automation.
- Built automated product and content workflows integrating **Shopify** and **Printful** APIs, reducing manual operations across catalog creation and fulfillment.
- Developed preprocessing pipelines for images and digital assets, including **validation**, **background removal**, **storage**, and downstream transformation.
- Integrated cloud storage and asynchronous processing to support reliable ingestion and scalable **product and marketing data flows**.

DeepMake

Apr. 2023 - Nov. 2024

Machine Learning / Data Engineer

Remote, USA

- Built scalable **Python** and **FastAPI** services for high-volume ML and data-processing workloads across training, inference, and media transformation pipelines.
- Developed production pipelines for **preprocessing**, **model inference**, **post-processing**, and artifact delivery for image and video workloads.
- Designed **AWS** infrastructure with dynamic compute allocation, autoscaling, monitoring, and logging for production reliability.
- Optimized latency, memory usage, and throughput for GPU-backed services and implemented **Sentry-based observability** to reduce failures.

CAREL USA

Jul. 2022 - Apr. 2023

Software Developer & Automation Engineer

Roswell, GA

- Built **data pipelines and dashboards** supporting energy analytics, forecasting, and operational visibility for HVAC systems.
- Processed **IoT telemetry data** used to analyze equipment behavior, performance, and efficiency trends.
- Integrated operational data sources, automation workflows, and **Structured Text PLC** logic to improve reporting and system efficiency.

SELECTED PROJECTS

Notar-IA – End-to-end legal document automation system built with **FastAPI**, **pdfplumber**, structured extraction, local RAG, validation/retries, and DOCX/PDF output generation for heterogeneous legal packets.

Creator Search 10M – Hybrid creator-brand retrieval design using **OpenSearch BM25 + HNSW**, cross-encoder

reranking, hard-constraint parsing, and offline **Precision@K / NDCG** evaluation over a 300K-profile simulation.

ED Triage Support Assistant – **Postgres**-backed triage console that ranked synthetic patients into explainable priority bands and simulated incoming vitals and notes for analytics-driven decision support.

COTABA Fleet Analytics Dashboard – Operational fleet analytics platform built with **FastAPI** and **React**, integrating live **Howen** telemetry through WebSocket ingestion and REST snapshots. Delivered 24-hour, weekly, and 30-day fleet metrics, authenticated dashboards, and admin diagnostics for anomalies, backfills, and auditability.

EDUCATION

Universidad de Los Andes (UNIANDES)	Bogotá, Colombia
M.Sc. in Civil Engineering (Geotechnical)	Jun. 2021
Universidad Pontificia Bolivariana (UPB)	Bucaramanga, Colombia
B.Sc. in Civil Engineering	Dec. 2017

SKILLS

- **Programming:** Python, SQL, JavaScript (Node.js), Bash
- **Data & Analytics:** ETL, data pipelines, data processing, Pandas, Polars, feature engineering, analytics, BigQuery
- **Databases & Storage:** PostgreSQL, BigQuery, OpenSearch, Supabase, Pinecone, Qdrant
- **Backend & Integration:** FastAPI, REST APIs, microservices, API integration, asynchronous processing
- **Cloud & Infrastructure:** AWS (EC2, S3, Lambda), GCP (Vertex AI, Compute Engine, Cloud Storage), Docker, CI/CD, Git, Linux/Unix
- **Evaluation & Reliability:** Precision@K, NDCG, data quality workflows, monitoring, observability, debug artifacts
- **Automation & ML:** n8n, serverless architectures, workflow automation, PyTorch, TensorFlow, Scikit-learn, ML pipelines